

Jason Huang

WEI-CHIEH HUANG

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EDUCATION

University of Illinois Chicago (Expected May 2028)

Ph.D. in Computer Science | Advisor: Prof. [Philip S. Yu](#) | GPA: 4.0/4.0

Aug. 2024 - Present

Chicago, IL, USA

Cornell University

M.S. in System Engineering | Major GPA: 3.92/4.0

Sep. 2020 – Dec. 2022

Ithaca, NY, USA

National Taiwan University of Science and Technology

B.S. in Chemical Engineering | Major GPA: 3.80/4.0

Sep. 2013 - Jun. 2017

Taipei, Taiwan

RESEARCH INTERESTS

- **Agentic Memory & Lifelong Personalization:** Memory substrates, context engineering, retrieval policies, self-evolving memory, long-context reasoning, and cross-domain user modeling [1, 2, 16, 17, 18].
- **Agent Evaluation & Interactive Benchmarking:** Long-horizon trajectories, user simulation, human participation, interruptibility, mid-task steering, and diagnostic evaluation [5, 10, 11, 19].
- **Agentic Deep Research, Tool Use & RL:** Web search, agentic RAG, long-horizon reasoning, reinforcement learning, self-evolving skills, verifier-guided optimization, and safety guardrails [3, 6, 7, 20, 21, 24].
- **Human-Agent Collaboration & Trustworthy Systems:** Human-in-the-loop feedback, multi-agent debate, heterogeneous coordination, interaction protocols, and personality-aware safety [4, 9, 12, 13, 15, 19, 22].

WORK EXPERIENCE

Amazon | Amazon Quick

Applied Scientist Intern

Sep. 2026 – Present

Seattle, WA, USA

- Build an end-to-end multi-agent software generation system for Amazon Quick, integrating task planning, code synthesis, tool orchestration, and execution to automate sandbox environment creation.
- Optimize tool selection and invocation, agent memory and retrieval, multi-agent communication topology, execution harnesses, and loop engineering through self-improving workflows and recursive self-improvement (RSI).

GEICO | Research & Innovation

Data Scientist

Jan. 2022 – Dec. 2023

Chevy Chase, MD, USA

- Built an automatic multimodal document-verification pipeline that fuses textual information and visual patterns via a BERT and CNN fusion model, achieving 98% overall accuracy for insurance customer verification.
- Designed and deployed an underwriting ML pipeline to detect potential insurance fraud, reducing the overall loss ratio by 3.5% and generating projected annual savings of \$60M.

PROFESSIONAL SERVICES & TECHNICAL SKILLS

Program Committee/Reviewer: ACL 2026, ACL Industry Track 2026, ICML 2026, NeurIPS 2026

Programming Languages: Python, SQL, C/C++, MATLAB, PHP, Hack, F#, LaTeX

Libraries: PyTorch, vLLM, SGLang, verl, TRL, LangChain, PyG, DGL, RecBole

PUBLICATIONS & PREPRINTS

* Equal contribution.

- [1] **Wei-Chieh Huang**, Weizhi Zhang, Yuchen Wu, Yankai Chen, ..., Philip S. Yu, Xue Liu, Aylin Caliskan. Harness the Memory: A Holistic Evaluation of Memory Substrates in Memory Agents. **NeurIPS 2026 (ED Track)** [Paper].
- [2] **Wei-Chieh Huang**, ..., Philip S. Yu, Kai Shu. A Survey of Agent Memory in the Second Half: Towards Self-Evolving and Long-Horizon Agents. **TMLR 2026 (Survey Certification, GitHub stars: 238)** [Paper] [GitHub].
- [3] **Wei-Chieh Huang**, Henry Peng Zou, Yaozu Wu, Dongyuan Li, Yankai Chen, et al. Deep Research with Open-Domain Evaluation and Multi-Stage Guardrails for Safety. **ACL 2026 Main (Poster)** [Paper].

- [4] **Wei-Chieh Huang**, Cornelia Caragea. MADIAVE: Multi-Agent Debate for Implicit Attribute Value Extraction. **EACL 2026 Findings** [\[Paper\]](#).
- [5] Weizhi Zhang*, **Wei-Chieh Huang***, Yueqing Liang*, Liwei Jiang, et al. JARVIS-Bench: Benchmarking Personal Intelligence Agents on Long-Horizon Real-User Daily Traces. **NeurIPS 2026 (ED Track)** [\[Paper\]](#).
- [6] Weizhi Zhang*, Yangning Li*, **Wei-Chieh Huang***, et al. From Web Search towards Agentic Deep Research: Incentivizing Search with Reasoning Agents. Under Review, 2026. [GitHub stars: 867](#) [\[Paper\]](#) [\[GitHub\]](#).
- [7] Yangning Li*, Weizhi Zhang*, Yuyao Yang, **Wei-Chieh Huang**, Yaozu Wu, et al. Towards Agentic RAG with Deep Reasoning: A Survey of RAG-Reasoning Systems in LLMs. **EMNLP 2025** [GitHub stars: 457](#) [\[Paper\]](#) [\[GitHub\]](#).
- [8] Chin-Yuan Yeh, **Wei-Chieh Huang**, Hsi-Wen Chen, Chen Wang, De-Nian Yang, ..., Philip S. Yu, Ming-Syan Chen. AMOS: Active Mining with Observable Steps for Agenda-Driven Review Knowledge Discovery. **KDD 2026** [\[Paper\]](#).
- [9] Henry Peng Zou*, **Wei-Chieh Huang***, Yaozu Wu*, Yankai Chen, Chunyu Miao, et al. LLM-Based Human-Agent Collaboration and Interaction Systems: A Survey. **ACL 2026 Findings** [GitHub stars: 237](#) [\[Paper\]](#) [\[GitHub\]](#).
- [10] Yaozu Wu*, **Wei-Chieh Huang***, Jizhou Guo*, Dongyuan Li, ..., Xue Liu, Philip S. Yu. HAS-Bench: Evaluating LLM-Based Human-Agent Systems under Configurable Human Participation. arXiv:2607.04329, 2026. [\[Paper\]](#).
- [11] Henry Peng Zou*, Chunyu Miao*, **Wei-Chieh Huang***, ..., Philip S. Yu. When Users Change Their Mind: Evaluating Interruptible Agents in Long-Horizon Web Navigation. arXiv:2604.00892, 2026. [\[Paper\]](#) [\[GitHub\]](#).
- [12] Yaozu Wu*, Dongyuan Li*, ..., **Wei-Chieh Huang**, et al. Multi-Agent Autonomous Driving Systems with Large Language Models: A Survey of Recent Advances, Resources, and Future Directions. **EMNLP 2025** [\[Paper\]](#).
- [13] Henry Peng Zou*, **Wei-Chieh Huang***, Yaozu Wu*, Chunyu Miao, Dongyuan Li, ..., Philip S. Yu. A Call for Collaborative Intelligence: Why Human-Agent Systems Should Precede AI Autonomy. Under Review, 2026. [\[Paper\]](#).
- [14] Yangning Li, Tingwei Lu, Yinghui Li, Yankai Chen, **Wei-Chieh Huang**, ..., Philip S. Yu. Teaching According to Talents! Instruction Tuning LLMs with Competence-Aware Curriculum Learning. **EMNLP 2025 Findings** [\[Paper\]](#).
- [15] Yaozu Wu, Jizhou Guo, Dongyuan Li, Henry Peng Zou, **Wei-Chieh Huang**, Yankai Chen, Zhen Wang, ..., Philip S. Yu. PSG-Agent: Personality-Aware Safety Guardrail for LLM-Based Agents. Under Review, 2026. [\[Paper\]](#).
- [16] Weizhi Zhang, Xiaokai Wei, **Wei-Chieh Huang**, et al. MemoryCD: Benchmarking Long-Context User Memory of LLM Agents for Lifelong Cross-Domain Personalization. **ICLR 2026 Lifelong Agent Workshop** [\[Paper\]](#).
- [17] Enze Ma, Yufan Zhou, **Wei-Chieh Huang**, Jie Yang, Huanhuan Ma, ..., Philip S. Yu, Zhen Wang. MemAudit: Auditing Long-Term Agent Memory via Hidden User-State Recovery. arXiv:2606.24595, 2026. [\[Paper\]](#).
- [18] Ruiyao Xu, Tiankai Yang, **Wei-Chieh Huang**. HyperSkill: Self-Evolving LLM Agents via Hypergraph-Structured Skill Memory. arXiv:2608.16114, 2026. [\[Paper\]](#).
- [19] Chunyu Miao, Henry Peng Zou, Yangning Li, Yankai Chen, ..., **Wei-Chieh Huang**, ..., Philip S. Yu. RECODE-H: A Benchmark for Research Code Development with Interactive Human Feedback. **ICLR 2026** [\[Paper\]](#) [\[GitHub\]](#).
- [20] Hanrong Zhang, Shicheng Fan, Henry Peng Zou, Yankai Chen, ..., **Wei-Chieh Huang**, ..., Philip S. Yu. CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification. **COLM 2026** [\[Paper\]](#).
- [21] Hanrong Zhang, Yankai Chen, Shicheng Fan, ..., **Wei-Chieh Huang**, ..., Dawn Song, Philip S. Yu, Xue Liu. Scaling LLM Agent Learning with Data Synthesis: A Comprehensive Survey. Under Review, 2026. [\[Paper\]](#).
- [22] Wooseong Yang, **Wei-Chieh Huang**, Weizhi Zhang, Yu Wang, Philip S. Yu, Junhyun Lee. XBridge: Entity-Grounded Latent Bridge for Heterogeneous LLM Communication. **NeurIPS 2026** [\[Paper\]](#).
- [23] Kening Zheng, **Wei-Chieh Huang**, Jiahao Huo, ..., Xuming Hu, Philip S. Yu. Unveiling Language Routing Isolation in Multilingual MoE Models for Interpretable Subnetwork Adaptation. **EMNLP 2026 Findings** [\[Paper\]](#).
- [24] Langzhou He, Junyou Zhu, Yue Zhou, ..., **Wei-Chieh Huang**, ..., Philip S. Yu, Qitian Wu. Resolving Action Bottleneck: Agentic Reinforcement Learning Informed by Token-Level Energy. arXiv:2605.14558, 2026. [\[Paper\]](#).
- [25] Liancheng Fang, Aiwei Liu, Henry Peng Zou, ..., **Wei-Chieh Huang**, Xue Liu, Philip S. Yu. Locally Confident, Globally Stuck: The Quality-Exploration Dilemma in Diffusion Language Models. **COLM 2026** [\[Paper\]](#).
- [26] Jiahao Huo, Yu Huang, Yibo Yan, ..., **Wei-Chieh Huang**, ..., Philip S. Yu. CausalEmbed: Auto-Regressive Multi-Vector Generation in Latent Space for Visual Document Embedding. Under Review, 2026. [\[Paper\]](#).
- [27] Chengze Li, Lingwei Wei, Li Sun, Hongbo Lv, Jie Yang, ..., **Wei-Chieh Huang**, Enze Ma, Philip S. Yu. ARC-STAR: Auditable Post-Hoc Correction for PDE Foundation Models. arXiv:2605.22222, 2026. [\[Paper\]](#).
- [28] **Wei-Chieh Huang**, Qianzhi Zhang, Fengqi You. Impacts of Battery Energy Storage Technologies and Renewable Integration on the Energy Transition in the New York State. **Advances in Applied Energy**, 2023. [\[Paper\]](#).

SELECTED AWARDS & HONORS

TMLR Survey Certification (first author), “A Survey of Agent Memory in the Second Half”
Advances in Applied Energy (2023) Highly Cited Research Papers Award

Aug. 2026
 Dec. 2023